

WS Tech – JSON with JavaScript Functions (Node.js Lab)

Task 1: Create JSON File

- 1 Create a file named students.json
- 2 Add at least 3 student records
- 3 Each student should have: name, age, course, marks
- 4 Ensure JSON keys use double quotes

Task 2: Read JSON in Node.js

- 1 Create app.js file
- 2 Use require('./students.json') to load JSON file
- 3 Print all student names using console.log()
- 4 Run using: node app.js

Task 3: Create Functions to Work with JSON Data

- 1 Create function getAllStudents() to print all records
- 2 Create function getStudentByName(name) to find student
- 3 Create function getAverageMarks() to calculate average marks
- 4 Call all functions from app.js

Task 4: Convert Object to JSON

- 1 Create a new JavaScript object for a student
- 2 Convert object to JSON using JSON.stringify()
- 3 Print the JSON string in console

Task 5: Write JSON File (Basic)

- 1 Use fs module
- 2 Use fs.writeFileSync() to save new JSON file
- 3 Store at least 2 student records
- 4 Verify file is created successfully

Bonus Task (Apply Level)

- 1 Use `fs.readFile()` (asynchronous) to read JSON file
- 2 Handle error properly in callback
- 3 Parse JSON string using `JSON.parse()`
- 4 Display total number of students

Submission Instructions:

Submit `students.json` and `app.js` inside a folder named: `WS_Tech_JSON_Node_StudentName`